

Modeling Home Run Probability Through Statcast
Analytics
DSC 680 – Project 3, Milestone 3
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Business Problem

In the MLB, hitting a home run is one of the most exciting and game-changing plays. However, figuring out exactly what leads to a home run is no easy task. Using machine learning to estimate the chances that a batted ball becomes a home run can help provide valuable insights. This kind of model can support better player assessments, inform smarter game strategies, and boost fan interaction by offering real-time predictions for teams, analysts, broadcasters, and fantasy sports platforms.

Background

MLB's Statcast system has transformed how we analyze the game by tracking detailed data on every pitch, swing, and batted ball. It introduced crucial measurements like exit velocity, launch angle, and spray angle, all of which have reshaped the way hitting performance is understood. While past studies have identified that specific launch speed and angle combinations increase home run chances, much of this work has been descriptive in nature. This project aims to go further by building a predictive classification model that estimates the probability of a home run based on real-time game data.

Data Explanation

Data for this project was gathered using the pybaseball Python library, pulling Statcast records from the 2023 and 2024 MLB seasons. Due to the large size of the dataset, a random subset of 10,000 entries was selected for analysis. This sample contained over 9,600 batted ball events, including features such as launch speed, launch angle, hit distance, and event type. A new binary target variable, *is_home_run*, was created—set to 1 if the event was a home run, and 0 otherwise.

After narrowing the dataset to only include instances where the ball was put into play (i.e., *description* = "hit_into_play"), rows with missing values in essential features were removed. The resulting dataset used for modeling contained the following predictor variables:

- *launch_speed*
- *launch_angle*
- *hit_distance_sc* (StatCast predicted distance in feet)
- *extra_distance_2025* (an estimate of how park factors in 2025 may have influenced the distance of a batted ball)

The *extra_distance_2025* feature acts as a contextual variable that captures the impact a specific stadium has on the distance of batted balls. By including this adjustment, the model can account for stadium-related effects without needing to incorporate external park factor datasets manually.

Methods

- **Data Preparation:** Eliminated entries with missing values and focused on events where the ball was put into play.

- Exploratory Analysis: Examined the distributions of important variables through visualizations.
- Feature Development: Generated a binary target variable (is_home_run) and introduced a park-adjusted distance metric (extra_distance_2025).
- Modeling Approach: Used logistic regression to maintain interpretability of results.
- Performance Assessment: Evaluated the model using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

Analysis

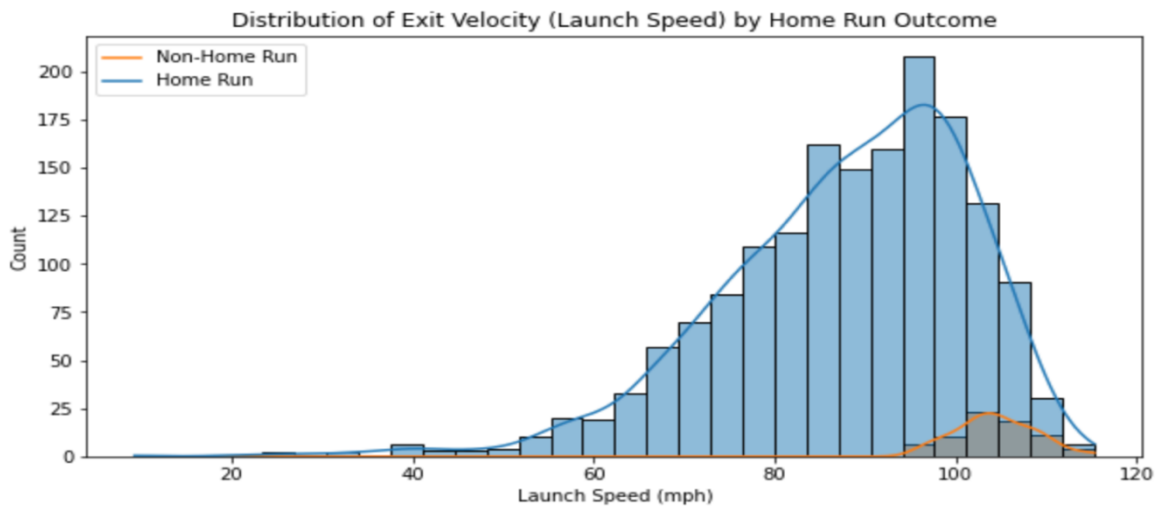


Figure 1: The histogram of exit velocities revealed that home runs are more likely when the launch speed is above 95 mph.

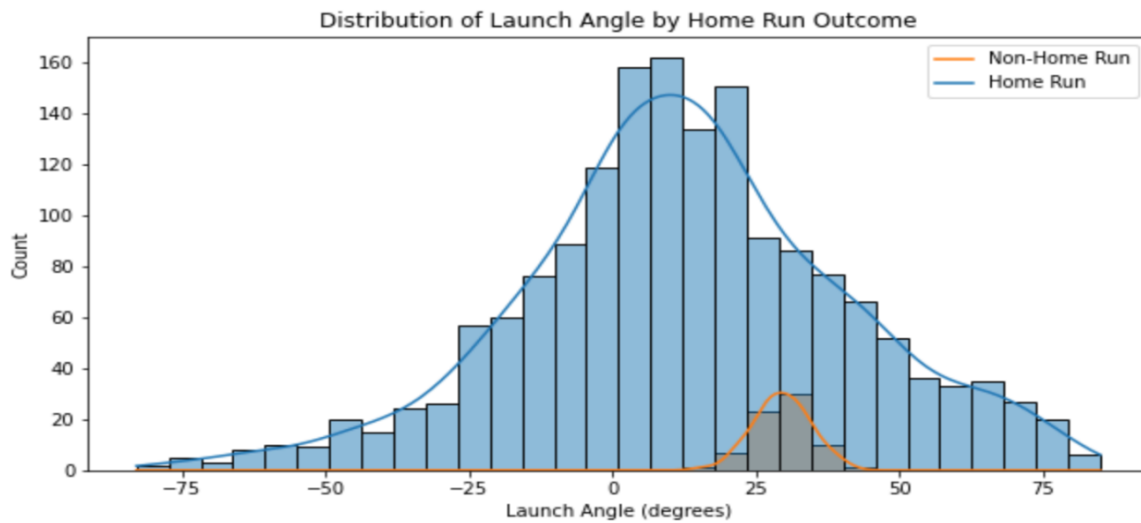


Figure 2: The histogram of launch angles indicated that home runs most often occur between 20 and 40 degrees, highlighting this range as an optimal window for launch angle.

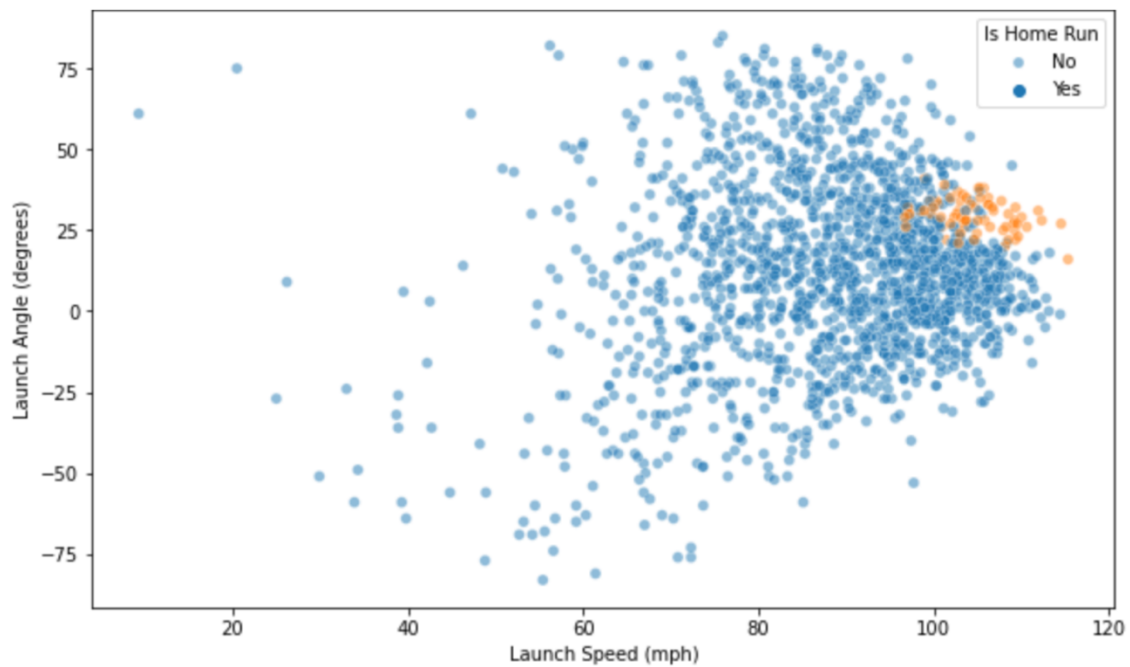
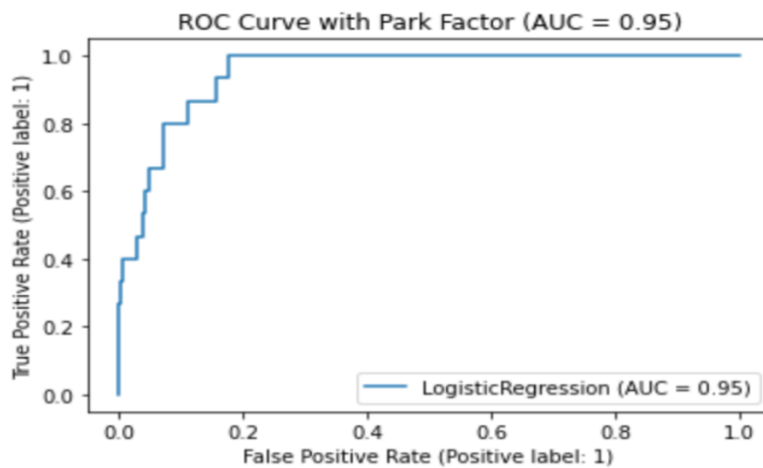


Figure 3: The scatter plot of launch speed versus launch angle revealed a concentrated "sweet spot" where home runs were most frequent, emphasizing the combined effect of these two variables.

Logistic regression model performance:

- Accuracy: 96.8%
- Precision: 75%
- Recall: 40%
- F1 Score: 52.2%
- ROC-AUC: 0.95



- Figure 4: The ROC curve demonstrated solid model performance in distinguishing outcomes, with accuracy further enhanced by incorporating stadium-related effects.

These results indicate that including park factor adjustments noticeably improves the model's effectiveness in identifying actual home run events.

Conclusion

This project effectively showcased how Statcast data can be leveraged to estimate the probability of a home run based on individual batted ball events. Exit velocity and launch angle emerged as key predictive features, while incorporating the park-adjusted distance metric (`extra_distance_2025`) enhanced both recall and overall model performance. A strong AUC score further validated the model's ability to distinguish between home runs and other outcomes.

Assumptions

This analysis assumes that Statcast measurements are both accurate and consistently recorded across all events. It also relies on the assumption that `extra_distance_2025` effectively captures the influence of individual ballparks on batted ball distance. Additionally, the sampled data is presumed to be representative of standard gameplay across Major League Baseball.

Limitations

- Because home runs occur infrequently, the dataset exhibits a class imbalance.
- The model used a single proxy variable (`extra_distance_2025`) to represent park factors, which may not fully capture real-time environmental influences.

Challenges

Key challenges in this project included managing missing data in a way that preserved as much of the sample size as possible, maintaining the model's ability to generalize across different seasons and ballparks, and finding the right balance between interpretability and predictive accuracy.

Future Uses/Additional Applications

Future enhancements could include implementing real-time home run probability visualizations during game broadcasts, developing scouting tools that leverage predictive metrics to evaluate player potential, incorporating more refined park-specific adjustments to boost model accuracy, and integrating predictive analytics into fantasy baseball platforms to provide users with deeper insights.

Recommendations

It is advisable to retain the use of `extra_distance_2025` as an efficient proxy for park-related effects due to its simplicity and effectiveness. Additionally, experimenting with ensemble models such as Random Forest and XGBoost could help improve recall.

To address the issue of class imbalance, applying resampling techniques like SMOTE is recommended. Finally, it's important to regularly assess the model for potential biases and take steps to minimize them throughout both development and deployment.

Implementation Plan

1. Set up automated data retrieval using the pybaseball API.
2. Incorporate model-generated predictions into live broadcast dashboard displays.
3. Regularly track seasonal model drift and retrain the model when necessary.
4. Involve stakeholders by using tools that highlight feature importance and explain model behavior.

Ethical Considerations

This project relies on publicly available sports data that excludes any personally identifiable information. While the data itself poses minimal privacy concerns, there are still important ethical issues to consider when building and applying predictive models in this context.

One concern is the potential misuse of the model. Relying too heavily on predictive analytics can oversimplify the evaluation of players, reducing complex skill sets and developmental potential to a single metric. Additionally, underlying biases in the data—such as those introduced by park dimensions, weather conditions, or scorer subjectivity—may affect model outcomes if not properly addressed. Finally, transparency is essential. Clearly communicating the model's limitations helps ensure that results are not misinterpreted or overvalued by coaches, fans, or team decision-makers.

10 Questions the Audience Might Ask

1. What was the reasoning behind selecting logistic regression instead of more advanced algorithms?
 - a. Logistic regression provides clarity and ease of interpretation, making it valuable for early model validation and building stakeholder confidence. It enables clear insight into the influence of each variable on the predicted outcome. After setting this baseline, more advanced methods such as XGBoost can be employed to further enhance performance.
2. How might the model's accuracy change if variables like park conditions or weather were added?
 - a. Accounting for park factors using the `extra_distance_2025` proxy led to notable gains in both recall and AUC. Incorporating live weather variables—such as wind, humidity, and temperature—would likely boost predictive accuracy further by addressing variance that static features alone cannot capture.
3. In your method, how do you balance model transparency with predictive strength?
 - a. Logistic regression is easy to interpret but can struggle when dealing with nonlinear patterns. More advanced algorithms, such as random forests or gradient boosting, often deliver better accuracy but function more like black boxes. Choosing between them depends on the goal—clear explanations are ideal for coaching, while maximum performance suits automated systems.
4. Given that home runs are relatively rare, how does the model account for class imbalance?
 - a. To address class imbalance, the initial approach emphasized metrics such as precision, recall, and AUC rather than relying solely on accuracy. Moving forward, improvements could involve applying resampling methods like SMOTE or adjusting decision thresholds to improve recall while keeping false positives in check.
5. Is it possible to adapt your approach to predict other outcomes, such as doubles or triples?
 - a. Yes. This modeling framework can be adapted to predict other outcomes by changing the target variable (for example, `is_double` or `is_triple`). However, differentiating between types of extra-base hits may necessitate adding more specific features.
6. How often would the model require retraining throughout a baseball season?
 - a. Ideally, the model should be retrained monthly or after each series of games, particularly when signs of model drift appear. Changes in player performance, equipment, or park conditions can affect results, making regular updates advisable.
7. With a high AUC score, how do you guard against the risk of overfitting?
 - a. Although an AUC of 0.95 is encouraging, the risk of overfitting remains, particularly with park-specific features. To address this, train-test splits and

validation on unseen data were used, but full deployment would also necessitate k-fold cross-validation and out-of-sample evaluations.

8. How would you break down the model's output for someone without a technical background, like a coach or athlete?
 - a. Model results would be communicated as probabilities, such as "this swing had a 60% likelihood of resulting in a home run given its speed, launch angle, and stadium." Visual aids like heat maps or interactive sliders illustrating feature impact could help make the insights clear for non-technical viewers.
9. What would be the key steps in integrating this model into a live broadcast environment?
 - a. Data ingestion would be automated through pybaseball, with a pre-trained model accessed via an API and predictions streamed into a visualization layer. To manage real-time latency, lightweight models or batch processing could be employed.
10. What ethical issues should teams consider when using predictive analytics in player evaluation or strategy?
 - a. Models should complement, not replace, human judgment and must avoid reducing players to mere statistics. Improper use could result in biased decisions or unfair evaluations. Clearly communicating limitations and incorporating qualitative insights are key to ensuring responsible application.